



Module : Data Analytics & Algorithms

Weekly E-Log Notes

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Sustainability and Resilience Considerations

AI-Powered ITSM Intelligence Platform

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Week 1 – Project Research and Idea Discussion

This week we focused mainly on understanding the assignment requirements and discussing different project ideas. Initially, we explored domains like fintech, healthcare systems, AI automation, and enterprise analytics. After several discussions, we became more interested in solving real problems faced by IT support teams, especially incident management and SLA handling.

VENKATESWARA spent more time researching enterprise AI systems and machine learning possibilities for operational data. VAMSI KRISHNA focused on understanding assignment expectations, sustainability requirements, and how the project could eventually become a complete web-based platform instead of just ML notebooks.

We also discussed how AI could improve operational sustainability by reducing repeated manual work and how predictive systems could improve resilience during high incident volumes.

Challenges

- Difficulty finalizing one project idea because multiple domains looked interesting.
- No dataset access yet, so implementation work could not begin.
- We were unsure whether to keep the project small or build a larger platform.

What Went Well

- Finalized ITSM as the project direction.
- Identified several possible AI features for future implementation.
- Understood early that sustainability and resilience needed to be integrated naturally into the project.

What Didn't Go Well

- Progress felt slow because actual development had not started yet.
- Too many early ideas created some confusion initially.

Next Plan

- Wait for dataset access from the company.
- Continue researching technologies and architecture ideas.
- Start planning the ML pipeline.

Week 2 – Architecture Planning and Technology Research

Since the dataset was still pending, this week was mainly spent planning the architecture and researching technologies.

At first, we thought of building only ML models, but gradually we realized the project would look much stronger as a complete AI-powered web application. This changed the overall direction of the project significantly.

VENKATESWARA researched FastAPI, machine learning workflows, and backend integration possibilities. VAMSI KRISHNA focused more on Streamlit dashboard development and frontend design ideas.

We discussed adding features such as:

- SLA breach prediction
- Root cause prediction
- Incident classification
- Similar incident recommendation
- AI chatbot assistance
- Anomaly detection

At this stage, the project started evolving from a simple ML idea into a larger operational intelligence platform.

Challenges

- Dataset delay slowed implementation progress.
- It was difficult to visualize how all modules would connect together.
- We were still learning how Streamlit and FastAPI integration would work.

What Went Well

- Finalized Streamlit and FastAPI as core technologies.
- Planned several AI features early in development.
- Started understanding the bigger vision of the platform.

What Didn't Go Well

- Some architecture ideas became unnecessarily complicated initially.
- We changed our design direction multiple times.

Next Plan

- Begin dataset exploration once access is available.
- Start preprocessing and feature analysis.
- Identify suitable ML models.

Week 3 – Dataset Exploration and Understanding

This week we finally received the real-world enterprise incident dataset. Since the dataset contained confidential operational information, we decided not to expose the dataset publicly.

This week marked the actual beginning of development work.

VENKATESWARA mainly focused on understanding relationships between incident fields and exploring prediction possibilities. VAMSI KRISHNA focused more on preprocessing support, organizing feature understanding notes, and helping structure the cleaning workflow.

We explored:

- incident descriptions
- SLA details
- priority and urgency
- root cause fields
- resolution history

Initially, we thought the project would mainly focus on SLA prediction, but after exploring the dataset more deeply, we realized it could support several intelligent modules.

Challenges

- The dataset was more complex than expected.
- Some operational fields were difficult to interpret.
- Understanding relationships between incident attributes required additional analysis.

What Went Well

- Successfully explored the dataset structure.
- Identified multiple ML opportunities from the same dataset.
- Started building preprocessing workflows properly.

What Didn't Go Well

- Data understanding took longer than expected.
- Some early assumptions about the dataset turned out to be incorrect.

Next Plan

- Start ML model development.
- Perform feature engineering.
- Experiment with SLA prediction models.

Week 4 – Initial Machine Learning Development

This week we started building the first machine learning models for the platform.

VENKATESWARA mainly handled model experimentation and algorithm testing. VAMSI KRISHNA supported preprocessing, testing, and documenting observations during early experiments.

We worked on:

- SLA Breach Prediction

- Root Cause Prediction
- Incident Classification

We explored:

- Logistic Regression
- Random Forest
- Gradient Boosting
- XGBoost

After several experiments, boosting-based approaches performed better for SLA prediction because they handled feature interactions more effectively.

We also worked on:

- label encoding
- preprocessing
- categorical transformations
- feature alignment

This week was important because the project finally became a working AI system instead of only a planned idea.

We also discussed how SLA prediction could improve operational resilience by helping teams identify high-risk incidents before delays occurred.

Challenges

- Selecting the best algorithm was difficult initially.
- Several preprocessing changes affected prediction consistency.
- Some early model outputs were unstable.

What Went Well

- First ML workflows became functional.
- XGBoost-based approaches showed promising results.
- Better understanding of feature importance was achieved.

What Didn't Go Well

- Some models underperformed during testing.
- We had to repeat preprocessing steps multiple times.

Next Plan

- Introduce NLP capabilities.
- Improve incident intelligence.
- Explore recommendation systems using historical incidents.

Week 5 – NLP and Intelligent Recommendation System

This week the project evolved significantly after introducing NLP capabilities.

We realized that incident descriptions contained valuable information that structured features alone could not fully utilize.

VENKATESWARA mainly worked on TF-IDF pipelines and similarity logic. VAMSI KRISHNA focused on testing outputs, refining preprocessing, and improving recommendation flow.

We implemented:

- TF-IDF vectorization
- NLP-based incident classification
- Similar incident matching
- Resolution recommendation system

Using cosine similarity, the system could compare new incidents with historical tickets and suggest previous resolutions.

This became one of the most practical features because it reduced repeated troubleshooting effort and improved operational sustainability for IT teams.

During this week, we also started discussing AI chatbot integration because we felt the platform still lacked natural user interaction.

Challenges

- NLP integration was more difficult than expected.
- Similarity outputs were inconsistent during initial testing.
- Text preprocessing required multiple refinements.

What Went Well

- Recommendation system became functional.
- Similar incident matching produced meaningful outputs.
- The platform started feeling more intelligent and practical.

What Didn't Go Well

- Some recommendations were too generic initially.
- Similarity quality was inconsistent during early testing.

Next Plan

- Integrate ML workflows into the web application.
- Connect frontend and backend systems.
- Improve overall user interaction.

Week 6 – Web Application Development and FastAPI Integration

This week focused heavily on converting the ML workflows into a complete web application.

Initially, we were not satisfied with the interface because it looked too basic and did not match the enterprise-style platform we imagined. Because of this, we redesigned several parts of the UI multiple times.

VENKATESWARA mainly worked on FastAPI integration and backend workflows. VAMSI KRISHNA focused more on Streamlit UI improvements, dashboard layouts, and frontend testing.

We integrated:

- SLA prediction
- Root cause prediction
- Recommendation system

into the web application successfully.

This was one of the most difficult weeks technically because frontend-backend communication issues occurred frequently.

Challenges

- FastAPI integration issues caused delays.
- Frontend-backend synchronization problems required repeated debugging.
- Some API routes failed repeatedly before stabilization.
- UI redesigns took more time than expected.

What Went Well

- Web application became functional.
- ML outputs displayed successfully inside the dashboard.
- The project started looking more professional and enterprise-oriented.

What Didn't Go Well

- Integration work became more complex than expected.
- Some UI layouts looked inconsistent initially.
- Several debugging loops slowed progress.

Next Plan

- Add AI chatbot functionality.
- Improve platform usability and interaction.
- Expand system intelligence further.

Week 7 – AI Chatbot Integration and Interaction Improvements

This week focused on improving user interaction by adding the AI chatbot called “Nebula”.

We felt the platform had strong ML capabilities, but interaction still felt limited. Because of this, we decided to introduce an AI assistant that could provide troubleshooting support and improve usability.

VENKATESWARA mainly worked on Gemini API integration and chatbot logic. VAMSI KRISHNA focused on chatbot UI improvements and testing interactions.

The chatbot was designed to:

- answer IT-related questions
- assist with incidents
- provide troubleshooting suggestions

Several refinements were required because some early responses were too generic.

During chatbot integration, we also faced Gemini API configuration issues. Initially, we tried hiding the API key more securely, but the chatbot stopped working properly. After several debugging attempts, a new API key was generated and directly configured temporarily for stable testing and academic demonstration purposes.

We also improved the recommendation engine during this week so the chatbot and similarity system could work together more effectively.

Challenges

- Chatbot response consistency needed improvement.
- Gemini API configuration caused runtime issues.
- Debugging chatbot integration took significant time.
- UI adjustments were required for smoother conversational flow.

What Went Well

- Nebula chatbot became functional.
- User interaction improved significantly.
- Chatbot and recommendation engine started working together effectively.

What Didn't Go Well

- Some chatbot responses were too generic initially.
- API authentication handling caused repeated interruptions during testing.

Next Plan

- Add anomaly detection features.
- Explore forecasting and monitoring systems.

- Improve operational intelligence modules.

Week 8 – Forecasting, Anomaly Detection, and Monitoring

This week we expanded the platform with more advanced operational intelligence features.

We felt the platform was becoming reactive rather than proactive, so we decided to introduce forecasting and anomaly detection modules to improve resilience and operational awareness.

VENKATESWARA mainly worked on anomaly detection and forecasting logic. VAMSI KRISHNA focused on frontend testing, dashboard refinement, and visualization improvements.

We implemented:

- workload forecasting
- anomaly detection
- MLOps monitoring concepts

These features improved the resilience aspect of the platform because organizations could better prepare for high incident volumes and unusual operational behavior.

Challenges

- Integrating several advanced modules increased system complexity.
- Dashboard clarity became difficult to maintain.
- Debugging became more time-consuming due to interconnected modules.

What Went Well

- Forecasting and anomaly detection became functional.
- Platform became more enterprise-oriented and advanced.
- Monitoring concepts improved overall platform maturity.

What Didn't Go Well

- Some dashboard sections became overcrowded initially.
- Feature integration occasionally affected workflow stability.

Next Plan

- Improve advanced AI modules.
- Refine visualizations and workflows.
- Continue complete platform testing.

Week 9 – Advanced Features and Refinements

This week focused mainly on refining the platform and experimenting with additional AI concepts.

VENKATESWARA explored Agentic AI Swarm simulation and GNN-based root cause visualization concepts. VAMSI KRISHNA focused on improving dashboard responsiveness, testing workflows, and organizing the project structure.

At this stage, the project had evolved much further than we initially planned during the early weeks.

We also improved:

- dashboard design
- prediction displays
- workflow stability
- repository organization

Challenges

- Managing increasing project complexity became difficult.
- Experimental modules required additional refinement.
- Maintaining responsiveness while integrating more features required optimization.

What Went Well

- Platform became more polished and stable.
- Advanced AI concepts improved the innovation value of the project.
- Workflow became smoother and easier to demonstrate.

What Didn't Go Well

- Some experimental features became difficult to optimize fully.
- Additional debugging was still required for stable performance.

Next Plan

- Perform final testing and debugging.
- Prepare final project demonstration.
- Finalize documentation and deployment setup.

Week 10 – Final Testing and Project Completion

This week focused on final testing, debugging, and preparing the project demonstration.

Both team members worked collaboratively during this phase because most tasks involved complete system testing and final refinements.

We tested:

- new incident predictions
- chatbot functionality
- anomaly detection
- forecasting
- recommendation system

We also finalized:

- screenshots
- GitHub structure
- documentation
- presentation materials

The final web application successfully demonstrated:

- predictive analytics
- NLP
- AI-assisted troubleshooting
- forecasting
- anomaly monitoring
- intelligent recommendations

Looking back, the project evolved significantly from a simple ML idea into a much larger AI-powered ITSM intelligence platform.

Challenges

- End-to-end testing required significant coordination.
- Multiple interconnected modules made debugging more complex.
- Small UI and integration issues continued appearing during final testing.

What Went Well

- Complete platform became functional.
- Final demo worked successfully.
- All major AI features operated together properly.

What Didn't Go Well

- Final refinements took longer than expected.
- Some UI polishing tasks remained until the final stage.

Sustainability and Resilience Considerations

Your project mainly supports operational sustainability and enterprise resilience within IT incident management environments.

Sustainability

The platform improves sustainability by reducing repeated manual troubleshooting effort through AI-assisted automation and intelligent recommendations. Features such as incident classification, similarity-based resolution suggestions, and chatbot assistance help reduce time spent on repetitive operational tasks. This improves overall efficiency for IT support teams and allows resources to be utilized more effectively.

The recommendation engine also helps engineers reuse historical solutions instead of repeatedly solving similar incidents from scratch. This reduces operational overhead and supports long-term workflow sustainability within enterprise IT environments.

Resilience

The project improves resilience by helping organizations respond to incidents more proactively and efficiently. Features such as SLA breach prediction, anomaly detection, workload forecasting, and AI-assisted troubleshooting help identify high-risk situations earlier and support faster operational recovery.

Forecasting and anomaly detection modules also improve preparedness during periods of unusually high incident activity. The platform therefore supports more stable and resilient IT operations by enabling predictive and proactive decision-making instead of purely reactive incident handling.

Final Reflection

This project helped us understand the importance of iterative development, debugging, collaboration, and continuous improvement. Over time, the project evolved from a simple ML idea into a much larger AI-powered ITSM intelligence platform.

We gained practical experience in:

- Machine Learning
- NLP
- FastAPI
- Streamlit
- API integration
- debugging
- enterprise application development

The project also demonstrated how AI can improve operational sustainability and resilience by reducing manual effort, improving incident response efficiency, supporting proactive monitoring, and helping organizations manage operational workloads more effectively.